

School of Computer Applications and Technology

Mid Term Examination - Apr 2026

M.Sc. CS II Sem / MCA (All) II Sem / Int. BCA+MCA VIII Sem

Duration : 90 Minutes
 Max Marks : 50

Sem II - E1PY207T - Operating Systems

General Instructions

Answer to the specific question asked

Draw neat, labelled diagrams wherever necessary

Approved data hand books are allowed subject to verification by the Invigilator

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| 1) | List any four functions of an Operating System. | CO1(3) |
| 2) | Explain any three Operating System services. | CO1(5) |
| 3) | Explain the meaning of deadlock in an operating system. | CO2(5) |
| 4) | Apply scheduling criteria to choose the best algorithm. A system is designed for an interactive time-sharing environment where: 1. response time should be low 2. fairness is important 3. users should not wait too long for the first response Which scheduling algorithm is most suitable among: 1. FCFS 2. SJF 3. Round Robin 4. Priority Scheduling Justify your answer. | CO1(8) |
| 5) | Explain the role of long-term scheduler, short-term scheduler, and medium-term scheduler in process management. | CO1(8) |
| 6) | Analyze whether the following system is in deadlock using a Resource Allocation Graph (RAG). A system has four processes P1, P2, P3, P4 and four single-instance resources R1, R2, R3, R4. R1 is allocated to P1, and P1 is requesting R2 R2 is allocated to P2, and P2 is requesting R3 R3 is allocated to P3, and P3 is requesting R4 R4 is allocated to P4, and P4 is requesting R1 Analyze whether the system is in deadlock. Justify your answer. | CO2(9) |
| 7) | Evaluate Process vs Thread Switching Cost A server performs 25 context switches in 1 second. Process context switch cost = 4 ms Thread context switch cost = 1 ms Find: Total overhead if implemented using processes Total overhead if implemented using threads Evaluate which design is better for a web server | CO1(12) |

OR

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| For the following processes, compare FCFS and Non-Preemptive SJF scheduling. Process Arrival Time Burst Time P1 0 8 P2 1 4 P3 2 2 P4 3 1 Draw Gantt chart for both algorithms 1. Find WT and TAT 2. Evaluate which algorithm is better and justify | CO1(12) |
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